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Submission Summary

Conference Name
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases - Workshop Track

Track Name
[MLSA]13th Workshop on Machine Learning and Data Mining for Sports Analytics

Paper ID
466

Paper Title
An Empirical Analysis of Multi-Agent Systems for Football Data Analytics and Performance Prediction

Abstract
Recent advances in the reasoning capabilities of Large Language Models (LLMs), together with a rapid increase in available football data, create a new opportunity for agentic systems to provide data-driven insights for complex football queries such as match analysis, strategy, and outcome or league-rank prediction. Football analytics is dynamic and multi-player, which raises a central question: whether and when to use a Multi-Agent System (MAS) rather than a single LLM. Prior research suggests that multi-agent decomposition improves accuracy on complex, multi-step tasks. We test this in the football setting by comparing a MAS against a single LLM with identical tool access and a single LLM with only raw data access, isolating the effect of orchestration from the effect of tool access. We evaluate accuracy, token usage, and inference time on a ground-truth set of 50 Premier League queries. We find that the autonomous MAS variant with Claude Opus~4.8 performs best on out-of-coverage queries and use it for subsequent experiments. The MAS outperforms a single LLM with the same tool access (89\% vs.\ 83\%), and both outperform a single LLM with only raw data access (35\%). Structured database access is the dominant driver of accuracy, well above the gain from orchestration. We also show that adding a persistent memory layer yields clear cost savings on recurring queries but no accuracy benefit on independent ones, and that naive similarity-based caching is unsafe for temporal data, which motivates structured, parameter-aware reuse.

Created
2026-06-10 23:48:32 CEST

Last Modified

2026-06-10 23:48:32 CEST

Authors

Till Tornieporth (LSE) <t.tornieporth@lse.ac.uk>

Submission Files

LSE_SKY_MAS_FOOTBALL.pdf (1.6 Mb, 2026-06-10 23:48:23 CEST)